1 **(a)** 48 **[B1]**

(b) 56 **[B1]**

(c) $180 - 48 - 56 = 76$ **[B1]**

[Total Marks: 3]

2 **(a)** 125 **[B1]**

(b) Corresponding angles (are equal) **[B1]**

(c) 144 **[B1]**

[Total Marks: 3]

3 **(a)** Identifies alternate angles by drawing a parallel line through E OR forms a triangle by extending a line **[M1]**

85 **[A1]**

(b) $\frac{360}{24}$ **[M1]**

15 **[A1]**

[Total Marks: 4]

4 **(a)** 52 **[B1]**

(b) Angles on a straight line sum to 180°: $180 - (52 + 61)$ **[M1]**

67 **[A1]**

(c) Identifies angle $TVU = 61°$ (alternate angle to QTV) or angle $VTW + TVW$ exterior relationship **[M1]**

Angle $TVW = 180 - 61 = 119$ **[A1]**

[Total Marks: 5]

5 **(a)** $360 \div 24$ **[M1]**

15 **[A1]**

(b) $(15 - 2) \times 180$ **[M1]**

2340 **[A1]**

(c) Exterior angle $= 180 - 165 = 15°$ **[M1]**

$360 \div 15 = 24$ **[A1]**

[Total Marks: 6]

6 **(a)** $180 - (55 + 70)$ **[M1]**

55 **[A1]**

(b) 55 **[B1]**

(c) Angle $DAE = 55°$ (from part a) AND Angle $ADE = 55°$ **[B1]**

Two angles are equal, so the triangle is isosceles **[B1]**

[Total Marks: 5]

7 (a) Equates sum of angles to formula: $3 \times 140 + 165(n-3) = 180(n-2)$ **[M1]**

Expands brackets correctly: $420 + 165n - 495 = 180n - 360$ **[M1]**

Rearranges to isolate n: $15n = 285$ (or equivalent) **[M1]**

19 **[A1]**

(b) States $a + b + c = 180$ **[B1]**

Substitutes $a + b = c$ to get $c + c = 180 \implies 2c = 180 \implies c = 90$ completing the proof **[B1]**

(c) (i) Calculates exterior angle of $P = \dfrac{360}{12} = 30°$ so exterior angle of $Q = 15°$ **[M1]**

24 **[A1]**

(ii) Calculates interior angles of P and Q: $180 - 30 = 150°$ and $180 - 15 = 165°$ **[M1]**

$360 - (150 + 165) = 45$ **[A1]**

[Total Marks: 10]

8 (a) Sum of interior angles of a pentagon $= (5-2) \times 180° = 540°$ **[M1]**

$x + (x + 20) + (2x - 10) + (2x + 30) + 140 = 540$ **[M1]**

$6x + 180 = 540 \implies 6x = 360 \implies x = 60$ **[A1]**

(b) Substituting $x = 60$ into $(2x + 30)$ or checking all angle sizes $(60°, 80°, 110°, 150°, 140°)$ **[M1]**

150 **[A1]**

[Total Marks: 5]

9 (a) Recognises the sum of the interior angles of a pentagon is $540°$ **[B1]**

Sets up a correct equation: $(2x + 10) + 100 + 80 + 3x + (x + 50) = 540$ **[M1]**

50 **[A1]**

(b) Sets up a relationship for the angles, e.g., $4E + E = 180$ where E is the exterior angle **[M1]**

$E = 36°$ **[M1]**

10 **[A1]**

[Total Marks: 6]

10 (a) $5x + 10 = 7x - 30$ **[M1]**

$x = 20$ **[A1]**

Corresponding angles (are equal) **[B1]**

(b) Angle $GHD = 7(20) - 30 = 110°$ **[B1]**

Angle $GHK = 180 - 110 = 70°$ (Angles on a straight line) **[B1]**

Angle $GKH = 180 - (70 + 35) = 75°$ (Angles in a triangle) **[B1]**

[Total Marks: 6]

11 (a) (i) Equates the sum of the angles to the interior angle sum of a hexagon (720°):
$x + 2x + x + 30 + 2x + 10 + 3x - 20 + 160 = 720$ **[M1]**

Simplifies the equation to $9x + 180 = 720$ (or $9x = 540$) **[M1]**

60 **[A1]**

(ii) Sets up an equation for the exterior angle e:
$6.5e + e = 180°$ **[M1]**

$7.5e = 180° \implies e = 24°$ **[M1]**

Number of sides $n = \dfrac{360}{24} = 15$ **[A1]**

(iii) Let exterior angles be E_m and E_n. Sets up the first equation:
$E_m + E_n = 54$ **[B1]**

Relates interior difference to exterior difference:
$(180 - E_m) - (180 - E_n) = 18 \implies E_n - E_m = 18$ **[M1]**

Solves the simultaneous equations by addition (or substitution):
$2E_n = 72 \implies E_n = 36°$ **[M1]**

Calculates n: $n = \dfrac{360}{36} = 10$ **[A1]**

Calculates m: $E_m = 18° \implies m = \dfrac{360}{18} = 20$ **[A1]**

[Total Marks: 11]

12 (a) Recognises that $\angle PXQ = \angle BPX + \angle DQX$ (e.g. by drawing a parallel line through X to form alternate angles) **[M1]**

$(3x + 15) + (2x + 10) = 75$ **[M1]**

$5x + 25 = 75 \implies 5x = 50 \implies x = 10$ **[A1]**

(b) Expression for exterior or interior angle of P: $\dfrac{360}{n}$ or $180 - \dfrac{360}{n}$ **[B1]**

Expression for exterior or interior angle of Q: $\dfrac{360}{3n}$ or $180 - \dfrac{120}{n}$ **[B1]**

Forms a correct equation: $\left(180 - \dfrac{120}{n}\right) - \left(180 - \dfrac{360}{n}\right) = 16$ OR $\dfrac{360}{n} - \dfrac{120}{n} = 16$ **[M1]**

$\dfrac{240}{n} = 16 \implies n = 15$ **[A1]**

[Total Marks: 7]

13 **(a)** $\dfrac{(5-2)\times 180}{5}$ OR $180-\dfrac{360}{5}$ **[M1]**

108 **[A1]**

(b) Interior angle of hexagon $= 120°$ **[B1]**

Angle $AEI = 360 - 108 - 120 = 132$ **[A1]**

(c) $AE = EI$ (sides of regular polygons sharing an edge are equal) **[M1]**

Triangle AEI is isosceles **[M1]**

Angle $EAI = \dfrac{180-132}{2} = 24$ **[A1]**

[Total Marks: 7]

14 **(a)** $(5-2)\times 180 \div 5$ OR $180-(360\div 5)$ **[M1]**

108 **[A1]**

(b) Interior angle of square $= 90°$ **[B1]**

Interior angle of n-gon $= 360 - 90 - 108 = 162°$ **[M1]**

$n = 360 \div (180 - 162) = 20$ **[A1]**

(c) Recognises triangle PFA is isosceles (since $PA = PF$ from shared sides of regular polygons) **[M1]**

Angle $PFA = (180 - 162) \div 2 = 9$ **[A1]**

[Total Marks: 7]

15 **(a)** Finds an expression for the interior angle ABC using the sum of angles at a point:
Interior angle $ABC = 360° - (300 - 2x)° = (60 + 2x)°$ **[M1]**

Draws a parallel line through B and uses alternate angles to form an equation:
Interior angle $ABC = (3x + 10)° + (2x + 20)°$ **[M1]**

Equates the two expressions for the interior angle ABC:
$5x + 30 = 60 + 2x$ **[M1]**

Solves the equation correctly to show the result:
$3x = 30 \implies x = 10$ **[A1]**

(b) Calculates the interior angle ABC using $x = 10$:
Angle $ABC = 60 + 2(10) = 80°$ **[M1]**

Recognises triangle ABC is isosceles and sets up the base angle calculation:
$\dfrac{180-80}{2}$ **[M1]**

50 **[A1]**

[Total Marks: 7]